Syeda Tasnim Fabiha

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Education

University of Southern California (USC) **Aug 2022 – Present**
Ph.D. in Computer Science (in 05/27); M.S. in Computer Science, May 2025 (GPA: 3.71/4.0)
Institute of Information Technology, University of Dhaka **Dec 2017**
B.S. in Software Engineering (BSSE) (GPA: 3.88/4.0)

Technical Skills

Programming Languages: Python, C#, C, R, JavaScript, HTML/CSS
Frameworks & Libraries: ASP.NET, Angular, React, Bootstrap, Microsoft SharePoint, PyTorch, scikit-learn, TensorFlow, Pandas, NumPy
Databases: MySQL, SQL Server, MongoDB
AI/ML Tools & Techniques: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Large Language Models (LLMs)
Technologies: AWS, Git, Jira

Research and Industry Experience

Bloomberg LP — May 2026 – Aug 2026
CTO Intern — *New York, NY*
- Research and prototype agent-based AI systems using LLMs and MCP-style architectures to analyze cloud infrastructure, Terraform state, and application code for architecture understanding and modernization.
- Develop AI pipelines that identify software components, dependencies, and architectural deviations, generating actionable insights, real-time architecture diagrams, and technical documentation for complex distributed systems.
- Model cloud infrastructure and software relationships as dependency graphs to support architecture-compliant modernization, secure-by-design governance, audit readiness, and developer experience.

University of Southern California — Aug 2022 – Present
Research Assistant — *Los Angeles, CA*

Conducting research at the intersection of software engineering and machine learning, with a focus on modernizing legacy systems into service-based architecture and evaluating deep learning–based vulnerability detectors:
- Built a reinforcement learning based framework for method-level service identification from legacy systems, balancing cohesion–coupling with business alignment using a customizable reward using Python, and achieved 7–14% higher modularization quality and 18–22% stronger business capability alignment compared to two state-of-the-art techniques. (Work under review)
- Co-developed Vulnerability Identification via Perturbations framework to evaluate whether DL detectors learn true vulnerability features vs. spurious cues; results included in ICSE 2025.

Brain Station 23 — Jan 2017 – Jul 2022
Senior Software Engineer — *Dhaka, Bangladesh*
- Re-engineered the Intermodal Maintenance and Repair System (IMRS) for Kaleris, using **ASP.NET MVC, WCF REST, SQL Server, jQuery.** [imrs]
- Redesigned workflow, database schema, and existing features for IMRS, improving usability and performance by **20%**.
- Led a 3-engineer team and collaborated directly with the onshore team and stakeholders for developing to gather requirements, align deliverables, and ensure smooth cross-site communication.[imrs].
- Developed agency digitization solution for Metlife from scratch using **Web API, Angular 5, SQL Server**, streamlining recruitment process and improving onboarding time 60% faster. [Metlife AD]
- Automated build and deployment process using Azure DevOps CI/CD pipelines, ensuring faster and timely releases.
- Optimized SQL queries and refactored codebase, reducing initial load time to **70%** and improving performance by **30%**.
- Worked on improving the security of systems that cleared Veracode scans with zero critical/high findings, strengthening security compliance.

Awards, Certifications & Leadership

NSF Travel Awardee, ICSE Mentoring Workshop (2025).
USC Viterbi School of Engineering Ph.D. Fellowship (2022–2023)
PC Member, SVM 2026 @ ICSE 2026 ; ACM Peer Reviewer (cert)
VP, BDSA USC (2024); WINCC Organizer; Viterbi Mentor (cert).
Certified Azure Fundamentals (cert); HackerRank Angular (cert); SQL Gold (badges); Nat'l Hackathon 4th.